

The empirical recurrence for OEIS A231998, proved by transfer matrix

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Abstract

OEIS A231998 counts $2 \times (n+1)$ arrays over $\{0, \dots, 1\}$ subject to a condition imposed on every CELL over a neighbour set the entry names, and it carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. A cell's neighbourhood meets three consecutive columns, so the state of a transfer matrix is a pair of consecutive columns and each step settles the middle one; an array is a walk, and $a(n)$ is a walk count on $S = 20$ vertices, which is C -finite. Whether the conjectured recurrence annihilates it is then decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A231998 is “Number of $(1+1) \times (n+1)$ 0..1 arrays with no element having a strict majority of its horizontal and antidiagonal neighbors equal to one.”. It has offset 1 and begins

4, 6, 16, 39, 81, 168, 361, 780, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = a(n-1) + 3*a(n-3) + 3*a(n-4) + 3*a(n-5) + 3*a(n-6) - 2*a(n-8) - a(n-9)$.

As of the “Last modified” line on the live entry (Oct 01 2018 21:10:06, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 Cells, not blocks

The entry names the neighbours it means — horizontal is left and right, vertical is up and down, diagonal is northwest-to-southeast, antidiagonal is northeast-to-southwest, king-move is all eight — and every such set lies inside the 3×3 square centred on the cell. Here the set is $\{(-1, 0), (1, 0), (1, -1), (-1, 1)\}$, written as offsets (row, column) from the cell. Every cell carries the condition

no cell has a strict majority of its horizontal and antidiagonal neighbours equal to 1,

and the count of neighbours enters it, so the boundary is not a detail that can be papered over: a cell in the interior has 4 neighbours in this set and a cell on the boundary fewer, and a condition such as “a strict majority of its neighbours” therefore means different things at the two. In particular one cannot pad the array with a border of zeros — that would give every

cell the full 4 and change what is being counted. The outside is therefore carried explicitly, as a sentinel column above the first and below the last and as a sentinel column either side, and a neighbour that falls outside is simply absent from both counts.

Write an array as a sequence of columns $u_1, \dots, u_L$, each in $\{0, \dots, 1\}^2$. The neighbourhood of a cell in column u_i meets u_{i-1} , u_i and u_{i+1} and nothing else. Take as vertices the pairs $(r, s) \in (\{0, \dots, 1\}^2 \cup \{\emptyset\}) \times \{0, \dots, 1\}^2$, and put an edge $(r, s) \rightarrow (s, t)$ when column s satisfies the condition at every one of its 2 cells, given r above and t below. An array with L columns is then a walk of length $L - 1$ starting at $(\emptyset, u_1)$, each step settling one column, and the last column is settled by the terminal weight, which evaluates it with $\emptyset$ below. Hence, with M the adjacency matrix, ι the starting vector and τ the terminal weight,

$$a(n) = \iota^\top M^n \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of the conjectured recurrence. Then the residual $a(n) - \sum_i c_i a(n - i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j ; and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+9} - \sum_i c_i M^{j+9-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ satisfy the monic linear recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = a(n - 1) + 3a(n - 3) + 3a(n - 4) + 3a(n - 5) + 3a(n - 6) - 2a(n - 8) - a(n - 9)$$

for every $n > 9$.

Proof. The vector $w = q(M)\tau$ was formed by 9 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly. Those integers vanish from the index corresponding to $n = 9 + 1$ onwards, and the run was continued until the working vector was identically zero, which settles all larger j at once. By Lemma 1 the recurrence holds for every $n > 9$. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, the transfer model reproduces all 30 terms the entry publishes, exactly, in integer arithmetic. That check earned its place here. The neighbour offsets are named in the array's own frame — horizontal means along a row — but when the entry fixes the first dimension, as in “ $2 \times n$ ”, the walk runs along columns and the two components of every offset swap. Sets such as king-move, or horizontal together with vertical, are unchanged by that swap and hide the error completely; a set such as horizontal, diagonal and antidiagonal is not, and the counts came out as those of a different member of the same family until the offsets were transposed.

Second, the conjectured recurrence was evaluated directly on the entry's own published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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